LABORATORY REPORT

Experiment Title	Lab 1: Axial Tensile Testing and Strain Measurement Accuracy
Course / Subject	TSM2370 Materialteknikk
Experiment No.	
Date Performed	[DD/MM/YYYY]
Author / Student Name	[First Last Name]
Student ID	[Student ID]
Lab Partner(s)	[Lab Partner Names]
Instructor / TA	Rasmus Ullestad Relling

1. Abstract

[A concise summary (150–250 words) of the entire report. State the purpose, the methods used, the main results (with key numerical values), and the principal conclusion. Write the abstract last, after all other sections are complete.]

2. Introduction

2.1 Background and Theory

Engineering stress (σ) is the applied axial force (F) divided by the original cross-sectional area (A0). Engineering strain (ε) is the change in length (ΔL) divided by the original gauge length (L0). While the UTM load cell accurately measures force, calculating strain using crosshead travel introduces errors. An extensometer is required to directly measure elongation over the gauge length. The material progresses from linear elastic deformation into plastic deformation, characterized by strain hardening, necking, and fracture.

Relevant Formulas:

Engineering Stress: $\sigma = F / A_o$

Engineering Strain: $\varepsilon = \Delta L / L_o$

Young's Modulus: $E = \sigma / \varepsilon$

2.2 Objectives

Determine how an aluminum alloy (6061-T6 or 5052-H32) behaves under a controlled, axial pulling load until fracture. The experiment will also demonstrate the critical importance of using an extensometer compared to machine crosshead travel for accurate strain calculation.

2.3 Hypothesis

[State your hypothesis or expected outcome based on the theory here]

3. Materials and Equipment

Device / Chemical	Amount / Specs	Manufacturer	Notes
Universal Testing Machine	10 kN (Max 5kN chucks)	Lloyd (LR10K)	Used for axial pulling
Camera Extensometer	1 unit	Basler/Logitech	Optical strain measurement
Aluminum Sheet Dog-bones	1 mm thick, 12.5 mm width	ISO Standard	6061-T6 or 5052-H32
Digital Calipers	1 unit	-	For L0 measurement

4. Safety Considerations

4.1 Hazards

Pinch points in UTM grips, sharp edges on fractured aluminum specimens.

4.2 Personal Protective Equipment (PPE)

Safety glasses, closed-toe shoes, standard lab attire.

4.3 Waste Disposal

Safely dispose of the fractured aluminum pieces in designated metal recycling bins.

5. Methods / Procedure

Step 1: Take safety precautions and ensure the UTM safety shields are in place.

Step 2: Use digital calipers to measure the exact initial gauge length (L0) and width/thickness of the stamped aluminum part.

Step 3: Input the measured L0 into the extensometer software and clamp the specimen securely in the UTM grips.

Step 4: Initiate the tensile pull at the specified crosshead speed.

Step 5: Run the test until complete specimen fracture.

Step 6: Export the resulting raw force and elongation data from the LR10K software.

Step 7: Clean the workstation and safely dispose of the fractured aluminum pieces.

6. Results

6.1 Observations

[Document any qualitative observations made during the experiment here]

6.2 Data Table

Parameter / Measurement	Value	Unit
Yield Strength (Machine Travel)		MPa

Yield Strength (Extensometer)		MPa
Ultimate Tensile Strength (UTS)		MPa
Young's Modulus (Machine Travel)		GPa
Young's Modulus (Extensometer)		GPa

6.3 Calculations

[Show one full sample calculation for each type of computation (e.g., calculating Stress, Strain, or average Hardness). Include the equation, substituted values, and the result with correct units.]

6.4 Figures / Graphs

[Insert diagram or figure here (e.g., Stress-Strain curve or Hardness chart). Ensure axes are labeled with proper units.]

7. Discussion

7.1 Interpretation of Results

Discuss the variance in the linear elastic slope (E) between the two curves. Explain why the extensometer is mandatory for extracting true material properties.

7.2 Comparison with Theoretical Values

[Discuss how your experimental values compare with theoretical or literature values]

7.3 Sources of Error

Systematic errors: [Identify calibration or methodological errors]

Random errors: [Identify environmental or measurement scatter errors]

8. Conclusion

[Summarise the most important findings and answer the question posed in the objectives.]

9. References

- Häll, K. (2026). Materials Science & Engineering: Practical Characterization. University of South-Eastern Norway.
- USN Department of Science and Industry Systems (2026). TSM2370 Materialteknikk 2026 Course Material. University of South-Eastern Norway.

10. Appendices

Appendix A — Raw Data

[Insert raw data logs or printouts here]

Appendix B — Additional Figures

[Insert any additional photos or supplementary graphs here]

LABORATORY REPORT

Experiment Title	Lab 2: Thermal Processing and Quench Severity
Course / Subject	TSM2370 Materialteknikk
Experiment No.	
Date Performed	[DD/MM/YYYY]
Author / Student Name	[First Last Name]
Student ID	[Student ID]
Lab Partner(s)	[Lab Partner Names]
Instructor / TA	Rasmus Ullestad Relling

1. Abstract

[A concise summary (150–250 words) of the entire report. State the purpose, the methods used, the main results (with key numerical values), and the principal conclusion. Write the abstract last, after all other sections are complete.]

2. Introduction

2.1 Background and Theory

Hardness correlates strongly with a material's yield strength. Rapidly quenching steel in water prevents carbon diffusion, forcing the FCC austenite into a BCT lattice (martensite) – the hardest and most brittle structure. A faster quench (cold water) is more severe than a slower quench (hot water/oil), leading to different hardness levels. Reheating (tempering) the as-quenched martensite relieves internal stresses and allows carbon to precipitate into cementite particles, restoring toughness.

Relevant Formulas:

Vickers Hardness (HV): Calculated via indentation diagonal lengths (d1, d2) and applied test force.

2.2 Objectives

Measure localized resistance to plastic deformation across different microstructures of AISI 1045 steel. The experiment demonstrates how varying quench media (water vs. oil/hot water) and tempering processes dictate final mechanical properties.

2.3 Hypothesis

[State your hypothesis or expected outcome based on the theory here]

3. Materials and Equipment

Device / Chemical	Amount / Specs	Manufacturer	Notes
Medium-carbon steel blocks	15-25 mm width, 10-15 mm thick	AISI 1045	4 samples per group
Lab Furnaces	Capable of 1000°C	Tinius Olsen	Austenitizing and tempering
Quench Media Stations	20°C water, 70°C water, oil	-	For varying quench severity

Vickers Hardness Tester	1 unit	-	For measuring plastic deformation resistance
Polishing Equipment	As needed	-	To prepare surfaces for indentation

4. Safety Considerations

4.1 Hazards

Extreme heat (1000°C furnaces), hot steel blocks, risk of burns, quenching vapors and splashes.

4.2 Personal Protective Equipment (PPE)

Heat-resistant protective gloves, tongs, face shields, lab coat, closed-toe shoes.

4.3 Waste Disposal

Dispose of polishing waste properly; handle quenching fluids per lab environmental guidelines.

5. Methods / Procedure

Step 1: Take safety precautions and put on heat-resistant protective equipment (gloves, tongs, face shields).

Step 2: Student 1 (Sample A): Polish the untreated 1045 steel block and measure the baseline Vickers hardness.

Step 3: Student 2 (Sample B): Austenitize sample at 1000°C, hold for uniformity, then rapidly quench in 20°C cold water. Polish and measure hardness.

Step 4: Student 3 (Sample C): Austenitize sample at 1000°C, then quench in 70°C hot water OR quenching oil. Polish and measure hardness.

Step 5: Student 4 (Sample D): Austenitize at 1000°C and quench in 20°C water. Immediately transfer to the tempering furnace for the prescribed cycle. Air cool, polish, and measure hardness.

Step 6: Record all hardness indentations and clear the testing equipment.

6. Results

6.1 Observations

[Document any qualitative observations made during the experiment here]

6.2 Data Table

Parameter / Measurement	**Value**	**Unit**
Sample A Hardness (Baseline)		HV
Sample B Hardness (20°C Water Quench)		HV
Sample C Hardness (Reduced Severity Quench)		HV
Sample D Hardness (Tempered		HV

Martensite)		

6.3 Calculations

[Show one full sample calculation for each type of computation (e.g., calculating Stress, Strain, or average Hardness). Include the equation, substituted values, and the result with correct units.]

6.4 Figures / Graphs

[Insert diagram or figure here (e.g., Stress-Strain curve or Hardness chart). Ensure axes are labeled with proper units.]

7. Discussion

7.1 Interpretation of Results

Compare Sample B and Sample C to explain the concept of quench severity and cooling rates. Compare Sample B and Sample D to explain the microstructural effects of tempering, the relief of internal stresses, and the restoration of toughness.

7.2 Comparison with Theoretical Values

[Discuss how your experimental values compare with theoretical or literature values]

7.3 Sources of Error

Systematic errors: [Identify calibration or methodological errors]

Random errors: [Identify environmental or measurement scatter errors]

8. Conclusion

[Summarise the most important findings and answer the question posed in the objectives.]

9. References

- Häll, K. (2026). Materials Science & Engineering: Practical Characterization. University of South-Eastern Norway.
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Appendix A — Raw Data

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Appendix B — Additional Figures

[Insert any additional photos or supplementary graphs here]